

SAT Math

Nonlinear Equations and Systems 3

Question # ID

Answer

3.1 fc3d783a

6

In the xy -plane, a line with equation $2y = 4.5$ intersects a parabola at exactly one point. If the parabola has equation $y = -4x^2 + bx$, where b is a positive constant, what is the value of b ?

3.2 4661e2a9

A

$$\begin{aligned} x - y &= 1 \\ x + y &= x^2 - 3 \end{aligned}$$

Which ordered pair is a solution to the system of equations above?

A. $(1 + \sqrt{3}, \sqrt{3})$

B. $(\sqrt{3}, -\sqrt{3})$

C. $(1 + \sqrt{5}, \sqrt{5})$

D. $(\sqrt{5}, -1 + \sqrt{5})$

3.3 f65288e8

A

$$\frac{1}{x^2 + 10x + 25} = 4$$

If x is a solution to the given equation, which of the following is a possible value of $x + 5$?

A. $\frac{1}{2}$

B. $\frac{5}{2}$

C. $\frac{9}{2}$

D. $\frac{11}{2}$

3.4 f2f3fa00

5

During a 5-second time interval, the average acceleration a , in meters per second squared, of an object with an initial velocity of 12 meters per second is defined by the equation $a = \frac{v_f - 12}{5}$, where v_f is the final velocity of the object in meters per second. If the equation is rewritten in the form $v_f = xa + y$, where x and y are constants, what is the value of x ?

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Question #	ID		Answer
3.5	6ce95fc8	$2x^2 - 2 = 2x + 3$ Which of the following is a solution to the equation above? A. 2 B. $1 - \sqrt{11}$ C. $\frac{1}{2} + \sqrt{11}$ D. $\frac{1 + \sqrt{11}}{2}$	D
3.6	c303ad23	If $3x^2 - 18x - 15 = 0$, what is the value of $x^2 - 6x$?	5
3.7	7bd10ef3	$2x^2 - 4x = t$ In the equation above, t is a constant. If the equation has no real solutions, which of the following could be the value of t? A. -3 B. -1 C. 1 D. 3	A
3.8	66bce0c1	$\sqrt{2x+6} + 4 = x + 3$ What is the solution set of the equation above? A. $\{-1\}$ B. $\{5\}$ C. $\{-1, 5\}$ D. $\{0, -1, 5\}$	B

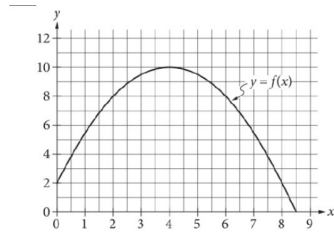
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Question # ID
3.9 97e50fa2

Answer

2, 8



The graph of the function f , defined by $f(x) = -\frac{1}{2}(x-4)^2 + 10$, is shown in the xy -plane above. If the function g (not shown) is defined by $g(x) = -x + 10$, what is one possible value of a such that $f(a) = g(a)$?

3.10 3d12b1e0

$$-16x^2 - 8x + c = 0$$

-1

In the given equation, c is a constant. The equation has exactly one solution. What is the value of c ?

3.11 71014fb1

$$(x-1)^2 = -4$$

D

How many distinct real solutions does the given equation have?

- A. Exactly one
- B. Exactly two
- C. Infinitely many
- D. Zero

3.12 e9349667

$$\begin{aligned} y &= x^2 + 2x + 1 \\ x + y + 1 &= 0 \end{aligned}$$

D

If (x_1, y_1) and (x_2, y_2) are the two solutions to the system of equations above, what is the value of $y_1 + y_2$?

- A. -3
- B. -2
- C. -1
- D. 1

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Question # ID
3.13 b03adde3

Answer

D

If $u - 3 = \frac{6}{t - 2}$, what is t
in terms of u ?

- A. $t = \frac{1}{u}$
- B. $t = \frac{2u + 9}{u}$
- C. $t = \frac{1}{u - 3}$
- D. $t = \frac{2u}{u - 3}$

3.14 30281058

A

In the xy -plane, the graph of $y = x^2 - 9$ intersects line p at $(1, a)$ and $(5, b)$,
where a and b are constants. What is the slope of line p ?

- A. 6
- B. 2
- C. -2
- D. -6

3.15 5910bfff

A

$$D = T - \frac{9}{25}(100 - H)$$

The formula above can be used to approximate the dew point D , in degrees Fahrenheit, given the temperature T , in degrees Fahrenheit, and the relative humidity of H percent, where $H > 50$. Which of the following expresses the relative humidity in terms of the temperature and the dew point?

- A. $H = \frac{25}{9}(D - T) + 100$
- B. $H = \frac{25}{9}(D - T) - 100$
- C. $H = \frac{25}{9}(D + T) + 100$
- D. $H = \frac{25}{9}(D + T) - 100$

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Question #	ID		Answer
3.16	1697ffcf	In the xy-plane, the graph of $y = 3x^2 - 14x$ intersects the graph of $y = x$ at the points $(0, 0)$ and (a, a). What is the value of a ?	5
3.17	ff2e5c76	$x^2 - 40x - 10 = 0$ What is the sum of the solutions to the given equation? A. 0 B. 5 C. 10 D. 40	D
3.18	2c5c22d0	$y = x^2 + 3x - 7$ $y - 5x + 8 = 0$ How many solutions are there to the system of equations above? A. There are exactly 4 solutions. B. There are exactly 2 solutions. C. There is exactly 1 solution. D. There are no solutions.	C
3.19	fc3dfa26	$\frac{4x^2}{x^2 - 9} - \frac{2x}{x + 3} = \frac{1}{x - 3}$ What value of x satisfies the equation above? A. -3 B. $-\frac{1}{2}$ C. $\frac{1}{2}$ D. 3	C

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Question # ID

Answer

3.20 58b109d4

-10

$$\begin{aligned}x^2 + y + 7 &= 7 \\ 20x + 100 - y &= 0\end{aligned}$$

The solution to the given system of equations is (x, y) . What is the value of x ?

3.21 7028c74f

31/3

$5(x + 7) = 15(x - 17)(x + 7)$ What is the sum of the solutions to the given equation?

3.22 e11294f9

31

The solutions to $x^2 + 6x + 7 = 0$ are r and s , where $r < s$. The solutions to $x^2 + 8x + 8 = 0$ are t and u , where $t < u$. The solutions to $x^2 + 14x + c = 0$, where c is a constant, are $r + t$ and $s + u$. What is the value of c ?

3.23 f5aa5040

20.25

In the xy -plane, a line with equation $2y = c$ for some constant c intersects a parabola at exactly one point. If the parabola has equation $y = -2x^2 + 9x$, what is the value of c ?

3.24 7dbd46d9

A

$$\begin{aligned}8x + y &= -11 \\ 2x^2 &= y + 341\end{aligned}$$

The graphs of the equations in the given system of equations intersect at the point (x, y) in the xy -plane. What is a possible value of x ?

- A. -15
- B. -11
- C. 2
- D. 8

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3.25 6011a3f8

Answer

A

$$64x^2 + bx + 25 = 0$$

In the given equation, b is a constant. For which of the following values of b will the equation have more than one real solution?

- A. -91
- B. -80
- C. 5
- D. 40

3.26 77c0cced

C

$$\begin{aligned} y &= 2x^2 - 21x + 64 \\ y &= 3x + a \end{aligned}$$

In the given system of equations, a is a constant. The graphs of the equations in the given system intersect at exactly one point, (x, y) , in the xy -plane. What is the value of x ?

- A. -8
- B. -6
- C. 6
- D. 8

3.27 3a9d60b2

9

$$2|4 - x| + 3|4 - x| = 25$$

What is the positive solution to the given equation?

3.28 abcd0012

4

$$(3x + p)(5x^2 - 45)(2x^2 - 16x + 6p) = 0$$

In the given equation, p is a positive constant. The sum of the solutions to the equation is $\frac{20}{3}$. What is the value of p ?

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3.29 88a0c425

$$-2x^2 + 20x + c = 0$$

In the given equation, c is a constant. The equation has exactly one solution. What is the value of c ?

- A. -68
- B. -50
- C. -32
- D. 0

Answer

B

3.30 722de804

$$(x - 47)^2 = 1$$

What is the sum of the solutions to the given equation?

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